

Sunilkumar Pathipati

Lead Engineer | Python · AI · Test Frameworks · AUTOSAR

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PROFESSIONAL SUMMARY

Lead Engineer with 9+ years of experience in automotive software engineering, embedded systems, artificial intelligence, and application development. Expertise in Python, C, AUTOSAR, machine learning, and Agile development practices. Experienced in developing safety-critical automotive solutions, including anti-lock braking systems for eBikes and two-wheelers, with strong exposure to ISO 26262 functional safety environments. Skilled in Python-based automation frameworks, local LLM tooling, intelligent testing platforms, software integration, verification and validation, release management, and engineering productivity improvement.

TECHNICAL SKILLS

Languages: Python, C, Embedded C, Bash, MATLAB

AI & Machine Learning: Artificial Intelligence, Machine Learning, LLMs, llama.cpp, Data Analytics

Automotive: AUTOSAR, ISO 26262, Android Infotainment, ABS Systems, Embedded Systems

Tools: Git, GitHub, GitHub Actions, JFrog Artifactory, Linux, Simulink

Core: Test Automation Frameworks, Software Validation, System Integration, Release Management

Methods: Agile Development, CI/CD Workflows, Software Development Life Cycle, Verification & Validation

Languages: English, Telugu, Hindi, Kannada, Tamil

PROFESSIONAL EXPERIENCE

Lead Engineer

September 2025 -- Present

BlueBinaries Engineering and Solutions Pvt Ltd (Client: Polestar)

Bengaluru, Karnataka, India

- Developing Python-based automation frameworks and local LLM solutions using llama.cpp for automotive infotainment testing and validation.
- Building AI-powered tools for test generation, log analysis, defect investigation, knowledge retrieval, and engineering productivity.
- Designing and maintaining GitHub Actions workflows, Git-based development processes, and automation utilities to streamline testing, build, and release activities.
- Leveraging Python, Bash scripting, and JFrog Artifactory to improve software quality, engineering efficiency, and validation workflows.
- Supporting secure on-premises AI tooling and local model deployment for automotive software testing environments.

Lead Engineer

January 2024 -- August 2025

Bosch Global Software Technologies

Bengaluru, India · Reutlingen, Germany

- Led automotive software development and validation for safety-critical embedded systems, including ISO 26262-compliant processes.
- Managed software integration, release management, verification, and validation across global programs.
- Led automation initiatives and engineering workflow improvements with international teams in Germany and India.
- Contributed to algorithm development, application development, and software release activities for automotive systems.

Specialist

January 2023 -- December 2023

Bosch Global Software Technologies

Bengaluru, Karnataka, India

- Developed and validated automotive software components for eBike and two-wheeler braking systems.
- Designed automation solutions and test frameworks to improve validation efficiency and software quality.

Senior Software Engineer

December 2021 -- January 2023

Bosch Global Software Technologies

Bengaluru, India · Reutlingen, Germany

- Worked on embedded software development, testing, debugging, and integration for automotive applications.
- Supported MATLAB/Simulink-based algorithm development and validation activities.
- Collaborated with international engineering teams in Germany on validation, automation, and software quality.

Software Engineer

July 2019 -- December 2021

Bosch Global Software Technologies

Bengaluru, Karnataka, India

- Developed embedded automotive software and supported testing activities for safety-critical systems.

- Worked on software verification, validation, debugging, and release support.

Software Engineer

March 2017 -- July 2019

HCL Technologies

Bengaluru, Karnataka, India

- Developed and tested embedded software solutions for automotive clients.
- Supported system integration, validation, debugging, and software testing activities.
- Worked with Linux-based development environments and automotive software tools.

PROJECTS

portfolio

Freelance

TypeScript

<https://github.com/psunil0001-sys/portfolio>

- Recruiter-facing site built from resume data and public GitHub projects. Live URL (after GitHub Pages is enabled): <https://psunil0001-sys.github.io/portfolio/> Open the URL Vite prints. Because this project is configured for GitHub Pages, the app is served at /portfolio/ (for example <http://localhost:5173/portfolio/>).

revops

Freelance

Python

<https://github.com/psunil0001-sys/revops>

- Interactive Streamlit app for tracking Kotak Multicap Fund - Regular Plan - Growth (scheme 149182), with a Manager-orchestrated multi-analyst research pipeline, Chroma memory, and a local LLM 60-day NAV scenario forecast.

LLM Junit4 Unittest Generator

Freelance

Python

<https://github.com/psunil0001-sys/LLM-Junit4-Unitest-Generator>

- UnitTestgen generates Kotlin/JUnit4 tests for Android Gradle modules using a local OpenAI-compatible llama.cpp server, verifies the result with Gradle/Kover through MCP, and repairs only generated test code until the selected Gradle tasks pass.

DAG example

Freelance

Python

<https://github.com/psunil0001-sys/DAG-example>

- Welcome to Astronomer! This project was generated after you ran 'astro dev init' using the Astronomer CLI. This readme describes the contents of the project, as well as how to run Apache Airflow on your local machine. Your Astro project contains the following files and folders:

calorietrack

Freelance

Dart

<https://github.com/psunil0001-sys/calorietrack>

- A modern Flutter-based mobile application utilizing the latest mobile development technologies and tools for building responsive cross-platform applications. - Android Studio / VS Code with Flutter extensions - Android SDK / Xcode (for iOS development)

EDUCATION

Bachelor of Technology in Electronics and Communications Engineering

2012 -- 2016

Ramireddy Subbaramireddy Engineering College

Andhra Pradesh, India

CERTIFICATIONS

- Certified Data Scientist
- Data Science Foundation
- Deep Learning with Keras and TensorFlow